1 Introduction and Objectives

This report details the steps taken to diagnose and resolve network malfunctions within the scope of the switching module. The primary objective of this lab is to restore inter-switch and intra-VLAN connectivity by identifying configuration errors on trunk links, incorrect port assignments, and protocol inconsistencies.
The key skills implemented throughout this process include:

- Analyzing switching tables and interface statuses.
- Detecting and correcting Native VLAN Mismatches.
- Configuring Switch Virtual Interfaces (SVIs) for infrastructure management.

2 Topology and Addressing Plan

The network architecture under scope consists of three interconnected Cisco switches serving 6 host computers distributed across different VLAN segments.

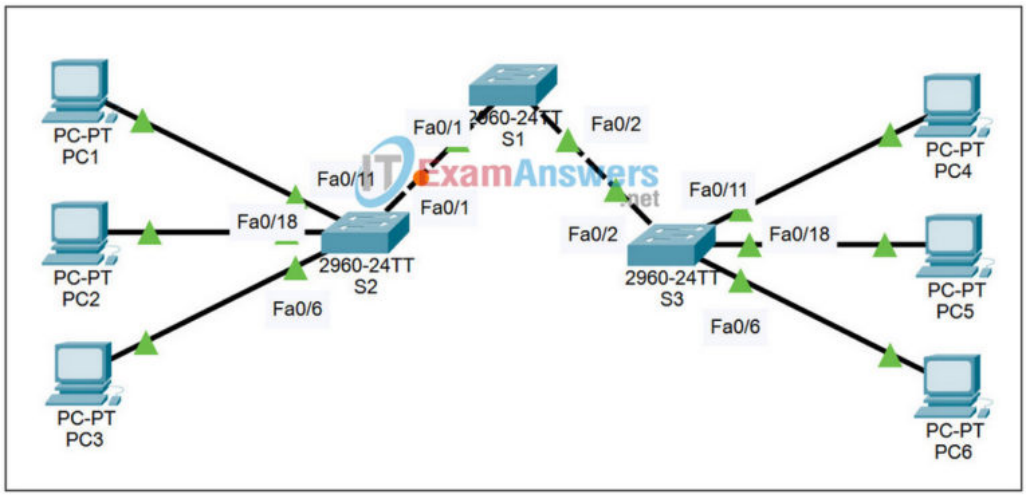


Figure 1: Cisco Network Topology

The network operates under the following VLAN allocation framework:

- **VLAN 10** : Faculty/Staff
- **VLAN 20** : Students
- **VLAN 30** : Guest(Default)
- **VLAN 56** : Management & Native (Infrastructure Management)

3 Troubleshooting Analysis (Initial State)

3.1 Initial Connectivity Test

Prior to any adjustments, a baseline series of ICMP requests (Pings) was performed between stations residing in the same VLAN. The outcome was definitive: all initial communication attempts failed completely.

Fire	Last Status	Source	Destination	Type	Color
	Failed	PC2	PC5	ICMP	
	Failed	PC1	PC4	ICMP	
	Failed	PC3	PC6	ICMP	
	Failed	PC2	PC3	ICMP	
	Failed	PC4	PC5	ICMP	

Figure 2: Systemic failure of intra-VLAN communication requests before correction.

3.2 Console Alerts and System Message Analysis

Upon opening the switch interfaces, the console immediately displayed recurring critical alerts generated by Cisco Discovery Protocol (CDP) and Spanning Tree Protocol (STP).

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with S3
FastEthernet0/2 (56).
%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id 56 on FastEthernet0/2
VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent local vlan.


%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with S3
FastEthernet0/2 (56).
```

Figure 3: Native VLAN Mismatch alerts and STP blocking state.

Technical Interpretation of System Log Errors:

1. `%CDP-4-NATIVE_VLAN_MISMATCH` : Interface `FastEthernet0/2` on switch S1 assumes the native VLAN is **VLAN 1**, whereas its neighbor S3 on the opposite end of the link uses **VLAN 56**. This desynchronization prevents untagged frames from being processed correctly.

2. `%SPANTREE-2-RECV_PVID_ERR` / `%SPANTREE-2-BLOCK_PVID_LOCAL` : To avoid potential switching loops resulting from this confusion, Spanning Tree immediately blocks interface `Fa0/2` on VLAN 1, neutralizing all transit traffic across this link.

3.3 VLAN Database and IP Addressing Verification

Executing the control command `show vlan br` on switch S1 revealed a major anomaly: VLANs 10, 20, and 30 required by the topology were entirely missing from its local database configuration.

```
S1#show vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
56   VLAN0056                          active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
```

Figure 4: Absence of user VLANs on the core switch S1.

On switch S2, an inspection of the logical addressing scheme exposed a critical administrative oversight: the dedicated management IP address (`192.168.56.12`) was mistakenly assigned to interface `Vlan1` instead of the proper SVI interface `Vlan56`, leaving the latter unassigned and inactive.

```
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  192.168.56.12   YES manual up                    up
Vlan56                 unassigned      YES manual administratively down down
S2#
```

Figure 5: IP addressing error on S2: Management IP mistakenly assigned to VLAN 1 instead of VLAN 56.

Concurrently, switch S3 retained the correct VLAN declarations, but its virtual interface tracking table indicated that the management IP address on VLAN 56 was not configured at all, showing a status of *Administratively Down*.

```
Vlan1                  unassigned      YES manual administratively down down
Vlan56                 unassigned      YES manual administratively down down
S2#
```

Figure 6: Unconfigured and deactivated VLAN 56 management interface on S3.

4 Resolution and Configuration Steps

To resolve the structural issues across the infrastructure, a systematic remediation strategy was executed sequentially on each network device.

4.1 Step 1: Creating Missing VLANs on S1

It is crucial to declare the missing user segments on S1 so it can forward the corresponding tagged traffic properly.

```
S1(config)# vlan 10
S1(config-vlan)# name Faculty/Staff
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# name Students
S1(config-vlan)# exit
S1(config)# vlan 30
S1(config-vlan)# name Guest(Default)
```

Listing 1: Creating and naming missing VLANs on S1

4.2 Step 2: Configuring Trunks and Native VLAN on S1

Interconnection links `Fa0/1` and `Fa0/2` must be explicitly configured as trunks, enforcing VLAN 56 as the native VLAN to permanently eliminate the CDP mismatch.

```
S1(config)# interface range Fa0/1 - 2
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# switchport trunk native vlan 56
S1(config-if-range)# switchport trunk allowed vlan 10,20,30,56
S1(config-if-range)# switchport nonegotiate
S1(config-if-range)# no shutdown
S1(config-if-range)# end
S1# write memory
```

Listing 2: Correcting trunk links on S1

4.3 Step 3: Access Port Assignment and Alignment on S2

On switch S2, user access interfaces are mapped into their respective target data segments, and infrastructure management interfaces are securely initialized.

```
S2(config)# interface range Fa0/2 - 5
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 56
S2(config-if-range)# no shutdown

S2(config)# interface range Fa0/11 - 17
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 10
S2(config-if-range)# no shutdown

S2(config)# interface range Fa0/6 - 10
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 30
S2(config-if-range)# no shutdown

S2(config)# interface range Fa0/18 - 24
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# no shutdown
S2(config-if-range)# end
S2# write memory
```

Listing 3: Configuring access and administration interfaces on S2

Next, the erroneous IP address layout on interface `Vlan1` of S2 is purged, shifting the management layer cleanly onto the SVI of `Vlan56`.

```
S2(config)# interface vlan 1
S2(config-if)# no ip address
S2(config-if)# exit
S2(config)# interface vlan 56
S2(config-if)# ip address 192.168.56.12 255.255.255.0
S2(config-if)# no shutdown
S2(config-if)# end
S2# write memory
```

Listing 4: Migrating management IP to the SVI VLAN 56 interface on S2

4.4 Step 4: Activating and Assigning Management IP on S3

Finally, on switch S3, the missing static logical parameters are specified for the VLAN 56 administration interface, forcing its operational state back to active using the `no shutdown` directive.

```
S3# configure terminal
S3(config)# interface vlan 56
S3(config-if)# ip address 192.168.56.13 255.255.255.0
S3(config-if)# no shutdown
```

Listing 5: Configuring Management SVI on S3

5 Verification and Validation (Final State)

5.1 Virtual Network Interface Status

Running the final validation diagnostic verifies that the management IP footprint has been successfully reassigned. The virtual interface `Vlan56` is now fully operational (`up / up`) with its intended addressing metrics, while interface `Vlan1` has been completely unassigned.

```
Vlan1                  unassigned      YES manual up
Vlan56                 192.168.56.12   YES manual up
```

Figure 7: Operational status of the SVI Vlan56 management interface after remediation.

5.2 Global Final Connectivity Tests

An end-to-end ICMP sweep confirms the thorough restoration of the local routing and switching framework:

- Hosts located within identical VLAN scopes achieve full connectivity across trunk corridors (e.g., PC1 to PC4, PC2 to PC5, PC3 to PC6).
- Broadcast domain boundaries are preserved accurately (requests from PC1 to PC2 or PC6 logistically fail, as expected without an active inter-VLAN layer).

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC2	PC5	ICMP	
	Successful	PC1	PC4	ICMP	
	Successful	PC3	PC6	ICMP	
	Failed	PC1	PC2	ICMP	
	Failed	PC1	PC6	ICMP	

Figure 8: Restored intra-VLAN connectivity and validated inter-VLAN isolation.

The enterprise segment has completely stabilized; Spanning Tree inconsistencies and CDP native mismatch triggers have been fully mitigated, yielding a final completion score of 100% within Packet Tracer.

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